Mourad Kraiem

Computer Science Engineering Student

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Profile

Final-year Computer Science Engineering student at ENSI seeking an end-of-study internship starting February 2027. Experienced in artificial intelligence, machine learning, computer vision, distributed AI, and full-stack development. Strong hands-on background through research internships, end-to-end AI projects, technical leadership, and national AI/ML competitions.

Education

Engineering Degree in Computer Science — 08/2024 – Present
National School of Computer Science (ENSI) — Manouba, Tunisia

Pre-Engineering Diploma in Mathematics and Physics — 08/2022 – 06/2024
Institut Préparatoire aux Études d'Ingénieurs de Monastir — Monastir, Tunisia

Ranked in the Top 10% in the national entrance exam for engineering schools.

Experience

Summer Internship – Explainable Graph-RAG for Arabic Medical QA — 07/2026 – 08/2026
LARODEC Laboratory — Tunisia

- Built an explainable Arabic medical QA system combining Neo4j knowledge graphs, multilingual embeddings, and hybrid lexical/vector/graph retrieval with provenance-aware evidence selection.
- Implemented query understanding, evidence assembly, claim-level verification, hallucination mitigation, abstention, and end-to-end retrieval/generation evaluation.

Summer Internship – Edge-Fog-Cloud Digital Twin and Federated Learning — 07/2026 – 08/2026
Research Internship in Smart Agriculture — Tunisia

- Developed the federated-learning component for yield prediction, with local model training performed on distributed fog nodes.
- Designed the training workflow in which fog-node updates were aggregated in the cloud and the resulting global model was redistributed within the edge-fog-cloud digital-twin architecture.

Summer Internship – Improving News Prediction through Multi-Agent Systems — 07/2025 – 08/2025
LARODEC Laboratory — Tunisia

- Built a multi-agent pipeline for Arabic health-news analysis using LLM-based extraction, entity normalization, symbolic graph reasoning, and GCN-based country-disease link prediction.
- Achieved an F1 score of 0.9848; the work was published in *Data & Knowledge Engineering* (Elsevier).

Projects

PCDent – AI-Powered Dental Assistance System — *Full-Stack AI / Computer Vision*

- Built a dentist-in-the-loop platform for panoramic dental radiograph analysis and clinical workflows using React, TypeScript, Django REST Framework, PostgreSQL, Redis/Celery, WebSockets, and Docker.
- Integrated YOLOv8m-seg for production dental detection and instance segmentation, with AI-assisted report generation and dentist validation.
- Designed and evaluated D-MODE, a research-oriented transformer architecture combining DINOv2, RT-DETR, and a mask-prediction head for dental detection and segmentation.

Penalty Kick ML Learning Path — *Machine Learning / Educational Project*

- Built a beginner-friendly ML/data-science learning path around predicting whether a football penalty is scored or missed, using a synthetic dataset to teach the full workflow step by step.
- Covered exploratory analysis, preprocessing, feature engineering, baseline and comparative modeling, validation, and evaluation in Python with Pandas, NumPy, scikit-learn, and notebooks.

Technical Skills

Programming Languages: Python, C, C++, Java, SQL, JavaScript, TypeScript
Machine Learning: PyTorch, TensorFlow, Scikit-learn, GNNs, Federated Learning, RAG
Tech Stack: Neo4j, Django REST Framework, FastAPI, PostgreSQL, Docker
Tools: Git/GitHub, Linux, Kaggle, VS Code, LaTeX

Awards & Leadership

- **5+ national AI/ML competition awards**, highlighted by 2nd Place at AI GOAT 1.0, where the team developed and evaluated deep-learning pipelines for image reconstruction and dense depth prediction, covering preprocessing, model training, and metric-based evaluation, and Project of the Year – Association Robotique ENSI for developing the ARE AI Chatbot.
- **Problem Setter – DataLeaders 2.0:** designed a beginner-friendly penalty-kick ML challenge, including the synthetic dataset, task formulation, baseline, evaluation metric, and Kaggle setup; reached 70 entrants, 28 teams, and 314 submissions.
- **Technical Manager – GODS 5.0:** managed the technical side of a 350-participant multi-label ESG text-classification competition and later developed an end-to-end transformer-based solution for the challenge.
- **Leadership & Competitive Programming:** Vice Chair, IEEE ENSI CIS Student Branch Chapter; Treasurer, CPS ENSI; Codeforces Specialist.